








Modelling Financial Markets

Chapter 3: Factor Models and Asset Pricing



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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **APT**: Arbitrage Pricing Theory
- **BAB**: Betting Against Beta
- **BET**: Bucharest Exchange Trading (index)
- **BH**: Benjamini–Hochberg (procedure)
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAGR**: Compound Annual Growth Rate
- **CAPM**: Capital Asset Pricing Model
- **CMA**: Conservative Minus Aggressive (investment factor)
- **CML**: Capital Market Line
- **ETF**: Exchange-Traded Fund
- **FDR**: False Discovery Rate
- **FF3**: Fama–French three-factor model
- **FF5**: Fama–French five-factor model
- **FWER**: Family-Wise Error Rate
- **GLS**: Generalised Least Squares
- **GMM**: Generalized Method of Moments
- **GRS**: Gibbons–Ross–Shanken (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HML**: High Minus Low (value factor)

Acronyms used in this material (2/2)

- **IPCA**: Instrumented Principal Component Analysis
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **MOM**: Momentum (factor)
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **OMV**: Österreichische Mineralölverwaltung — *Austrian Mineral Oil Administration (OMV group)*
- **PC1**: first Principal Component
- **PC2**: second Principal Component
- **PC3**: third Principal Component
- **PCA**: Principal Component Analysis
- **PRDS**: Positive Regression Dependence on a Subset (of p-values)
- **RF**: Risk-Free rate
- **RMSE**: Root Mean Squared Error
- **RMW**: Robust Minus Weak (profitability factor)
- **ROE**: Return On Equity
- **RP-PCA**: Risk-Premium Principal Component Analysis
- **SDF**: Stochastic Discount Factor
- **SMB**: Small Minus Big (size factor)
- **SML**: Security Market Line
- **UMD**: Up Minus Down (momentum factor)
- **US**: United States

Chapter Route and Outcomes

Four questions

- What explains expected returns? The CAPM and multifactor models
- How are exposures estimated? Betas, their uncertainty and shrinkage
- How are pricing restrictions tested? GRS, Fama–MacBeth, model comparison
- Do the findings survive? Multiple testing, latent factors, replication

After this chapter you can

- estimate a beta with HAC errors and shrink it
- test the CAPM alphas jointly and estimate risk premia with robust errors
- judge a new factor against multiple-testing hurdles and replication evidence

- **Part I**, Sections 1–5: 1 Route and data; 2 From Markowitz to the CAPM; 3 Estimating beta; 4 Testing the CAPM; 5 Multifactor models
- **Part II**, Sections 6–13: 6 Fama–MacBeth regressions; 7 Discount factors and model comparison; 8 Factor zoo and multiple testing; 9 Latent factors; 10 Factor investing; 11 Case study; 12 AI for scientific discovery; 13 Conclusions
- Technical details are linked from each topic; further study and code: ▸ Appendix

What Is a Factor Model?

- A **factor model** explains the return of asset i by a few common variables $f_{1,t}, \ldots, f_{K,t}$, the **factors**

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1} f_{1,t} + \cdots + \beta_{i,K} f_{K,t} + \varepsilon_{i,t}$$

- $R_{i,t}$: return of asset i in period t; $R_{f,t}$: risk-free rate; K: number of factors; α_i: average return not explained by the factors, 0 if they price the asset
- $\beta_{i,k}$: **factor loading** (exposure) of asset i to factor k; $\varepsilon_{i,t}$: asset-specific (idiosyncratic) shock
- Two uses in this chapter
  - Explain an asset: the technology ETF XLK moves 1.28 times the US market (Section 3)
  - Evaluate a fund: is its α_i positive once its exposures are priced (Sections 4 and 10)?
- Also: cost of capital, risk management (N assets need $N(N+1)/2$ covariances, K factors far fewer), portfolio construction (Campbell, Lo & MacKinlay (1997), ch. 5–6)
- Key message: expected returns should be explained by **exposures to priced risks**, not by the name of the asset

Data and Conventions

Series	Content	Period
Fama–French factors	market, size, value, profitability, investment	1963–2026
Momentum factor	winners minus losers	1963–2026
25 size × B/M portfolios	US stocks, two sorts	1963–2026
US sector ETFs	S&P 500 sectors: 9 from 1999, 11 from 2018	1999–2026
Factor ETFs	momentum, value, quality, min. volatility, size	2013–2026
BVB blue chips, BET	11 stocks and their index	2015–2026

- Sector ETF tickers: XLB materials, XLE energy, XLF financials, XLI industrials, XLK technology, XLP consumer staples, XLU utilities, XLV health care, XLY consumer discretionary; from 2018 also XLRE real estate and XLC communication services

- Factors and portfolios: Kenneth French Data Library, in percent (divide by 100); ETFs and stocks: daily adjusted closing prices (BET: closing values), to 18 Sep 2026
- **Excess return** $R^e_{i,t} = R_{i,t} - R_{f,t}$; $R_{f,t}$: one-month Treasury bill rate (RF in the French data); **Mkt-RF**: excess return of the value-weighted US stock market in the same data
- **B/M**: book value of equity over its market value; **BET**: the benchmark index of the BVB
- Simple returns for regressions on the French factors; log returns for rolling, BVB and PCA analyses
- BVB: daily $R_{f,t}$ treated as constant; it cancels in covariances, so betas are unchanged

Mean–Variance and the Tangency Portfolio

- Markowitz (1952): with weights w, expected excess returns $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, the frontier solves

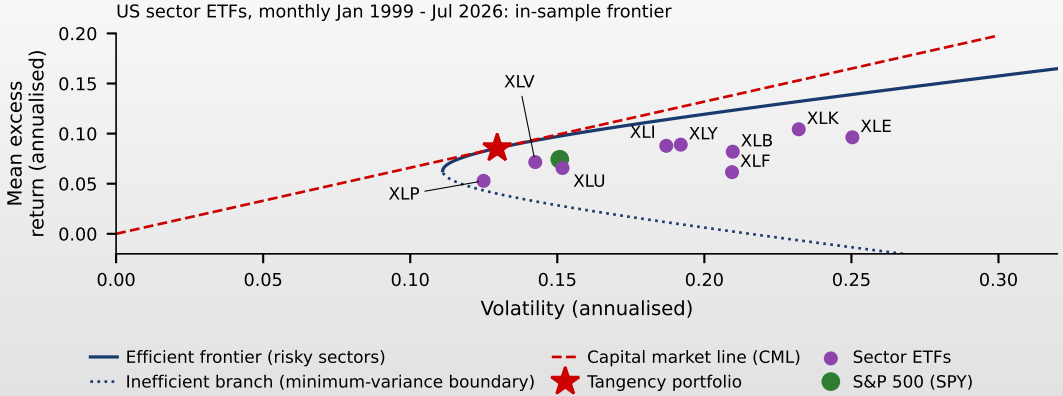
$$\min_{\mathrm{w}} \ \mathrm{w}^{\top}\boldsymbol{\Sigma}\,\mathrm{w} \quad \text{s.t.} \quad \mathrm{w}^{\top}\boldsymbol{\mu} = \mu^{*},\ \mathrm{w}^{\top}1 = 1$$

  - in words: the lowest variance for a target mean μ^{*}; 1: a vector of ones, so the weights sum to 1; $^{\top}$: transpose
- **Sharpe ratio** $SR_p = \mathbb{E}[R_p - R_f]/\sigma_p$; with a risk-free asset the best risky portfolio is the **tangency portfolio**

$$\mathrm{w}_{\mathbf{tan}} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} \,/\, 1^{\top}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$$

  - R_p, σ_p: return and volatility of portfolio p; the denominator rescales $\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$ so that the weights sum to 1
- **CML**: mixes of R_f and the tangency portfolio, $\mu_p = SR_{\mathbf{tan}}\,\sigma_p$; negative weights are **short positions**
- From the tangency portfolio to the beta pricing equation: Seminar 3, problem A2; estimation error and shrinkage: Chapter 4

The Efficient Frontier of US Sectors (1/2)



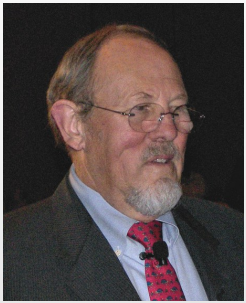
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The Efficient Frontier of US Sectors (2/2)

- Monthly excess returns of 9 sector ETFs, January 1999 – July 2026 ($T = 331$ months)
- Tangency portfolio: mean 8.55%, volatility 12.96%, Sharpe ratio 0.66
- SPY (the S&P 500 ETF, the market proxy): mean 7.44%, volatility 15.08%, Sharpe ratio 0.49: here the market is **not** on the frontier, although the CAPM requires it to be
- Tangency weights: -0.37 in financials, -0.21 in materials, $+0.37$ in health care
- The frontier is computed **in-sample**; estimated means are very noisy (Chapter 4)

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The CAPM: Sharpe, Lintner, Black



William F. Sharpe, Nobel Prize in Economics 1990

Photo: Larry D. Moore (2007); CC BY 4.0; Wikimedia Commons

- **CAPM**: Capital Asset Pricing Model, Sharpe (1964) and Lintner (1965)
- If every investor holds a mean–variance efficient portfolio, all hold the same tangency portfolio, which must then be the **market portfolio**
- Expected excess return is proportional to **beta**

$$\mathbb{E}[R_i] - R_f = \beta_i \left(\mathbb{E}[R_m] - R_f\right), \qquad \beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)}$$

- R_i, R_m: returns of asset i and of the market portfolio; $\mathbb{E}[R_m] - R_f$: **market risk premium**; β_i: sensitivity of asset i to the market, $\beta_i > 1$ amplifies market moves
- Black (1972): a version without a risk-free asset, the zero-beta CAPM (Section 4)
- Sharpe shared the 1990 Nobel Prize with Markowitz and Miller

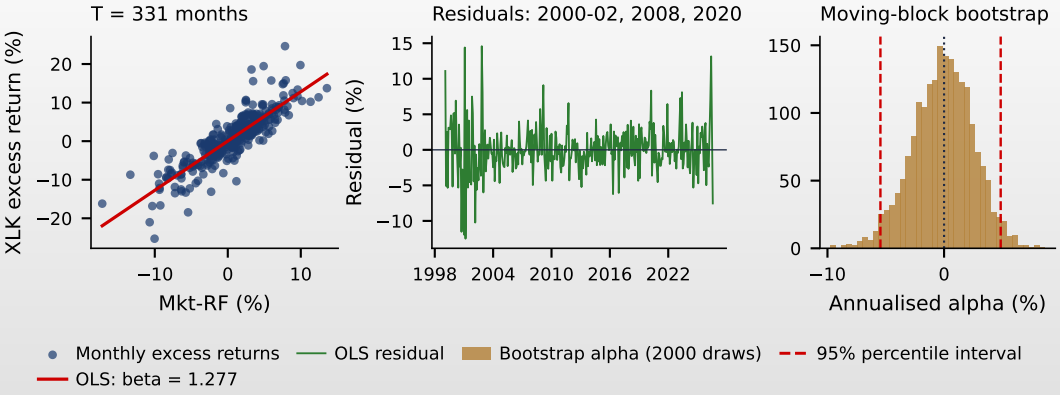
What the CAPM Implies

- Main assumptions
  - Mean–variance investors with identical beliefs, one period
  - Unlimited borrowing and lending at R_f; no taxes or trading costs
- Only beta is priced: no other characteristic should predict returns; idiosyncratic risk earns nothing
- **SML** (Security Market Line): the testable statement of the CAPM (Section 4)
  - every asset lies on the line (beta, mean), with slope equal to the market premium
  - the CML contains only efficient portfolios
- Cost of capital from the CAPM needs an estimated beta and an estimated premium: the error of the premium dominates
- Which portfolio is "the market"? The answer decides what a test of the CAPM can establish (Roll's critique, Section 4)

The Market Model, HAC Errors and Estimation Choices

- Market model $R^e_{i,t} = \alpha_i + \beta_i R^e_{m,t} + \varepsilon_{i,t}$; **OLS** (Ordinary Least Squares): $\hat{\beta}_i = \widehat{\mathrm{Cov}}(R^e_i, R^e_m)/\widehat{\mathrm{Var}}(R^e_m)$
- Variance split $\sigma_i^2 = \beta_i^2\sigma_m^2 + \sigma^2_{\varepsilon,i}$: systematic (priced under the CAPM) plus idiosyncratic (diversifiable); R^2 is the systematic share
  - σ_i^2, σ_m^2: variances of the asset and of the market; $\sigma^2_{\varepsilon,i}$: residual variance; $R^2 = \beta_i^2\sigma_m^2/\sigma_i^2$
- Volatility clustering (Chapter 1) makes residuals heteroskedastic: use **HAC** errors (Newey & West (1987))
  - Bartlett weights and $L = \lfloor 4(T/100)^{2/9} \rfloor$ lags (Newey & West (1994)); T: number of periods, $\lfloor \cdot \rfloor$: integer part
  - $T = 331$: $L = 5$; $T = 757$: $L = 6$; the point estimates are those of OLS
- Choices to report with every beta
  - Frequency and window: 3–5 years of monthly or 1–2 years of daily data; daily data suffer from asynchronous trading
  - Market proxy: SPY (S&P 500) or Mkt-RF for US assets, BET for Romanian stocks
  - Data cleaning: a scale error in the first records creates a fake return (DIGI: the closes of 17–22 May 2017 are about 725 times those from 23 May on; daily log return -6.6)

Worked Example: Beta of the Technology ETF (1/2)



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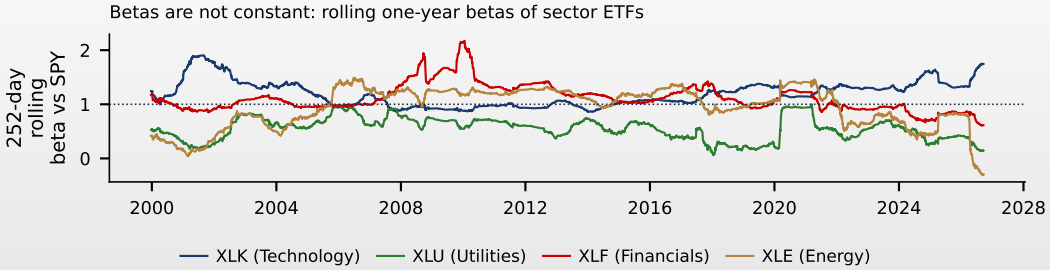
Worked Example: Beta of the Technology ETF (2/2)

		Standard error		
	Estimate	**Classical**	**White**	**Newey–West**
α (annual)	0.33%	2.27%	2.32%	2.45%
β	1.277	0.041	0.054	0.082

- XLK on Mkt-RF, monthly 1999–2026, $T = 331$, $R^2 = 0.742$
- **Classical**: errors assumed homoskedastic and serially uncorrelated; **White**: heteroskedasticity-consistent (White (1980)); **Newey–West**: also robust to autocorrelation, $L = 5$ lags
- Robust standard error (s.e.) of β: twice the classical one; heteroskedasticity explains part (White), serial correlation of $f_t\varepsilon_t$ the rest
- $t(\alpha) = 0.33/2.45 = 0.13$; bootstrap interval $[-5.5\%, 4.9\%]$

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Betas Are Not Constant



- OLS beta on the last 252 daily log returns, recomputed every day
- Financials (XLF): from 0.60 to 2.17; the peak follows the 2008–2009 banking crisis
- Technology (XLK): from 0.86 to 1.90; high in the dot-com years, 1.75 at the end
- Energy (XLE): from −0.31 to 1.49; negative at the end, when oil moved against the market
- Utilities (XLU): from 0.06 to 1.00, the classic defensive sector
- A full-sample beta hides these regimes; conditional models: Chapters 5–6

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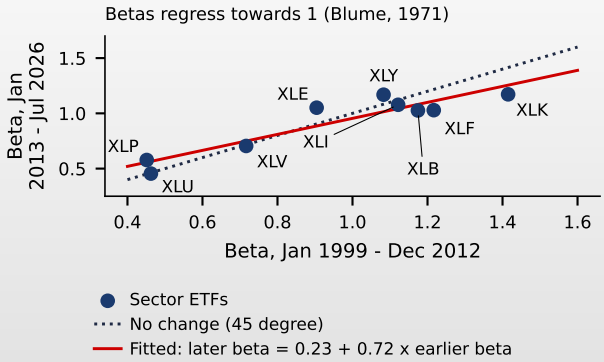
Shrinking Betas: Blume and Vasicek

- **Regression to the mean**: extreme estimated betas are partly noise and move towards 1 later
- Blume (1971): regress later betas on earlier betas; the practitioner rule is $\beta^{\text{Blume}} = 0.33 + 0.67\,\hat{\beta}$
- Vasicek (1973): Bayesian average of the estimate and the cross-sectional prior mean $\bar{\beta}$

$$\beta_i^V = w_i \hat{\beta}_i + (1 - w_i)\bar{\beta}, \qquad w_i = \frac{\sigma_\beta^2}{\sigma_\beta^2 + \text{se}(\hat{\beta}_i)^2}$$

- w_i: the weight of the estimate, larger when $\hat{\beta}_i$ is precise (small se); $\text{se}(\hat{\beta}_i)$: its standard error
- σ_β^2 is the variance of the **true** betas (Var_{cs}: variance across assets; $\overline{\text{se}^2}$: mean squared standard error): $\text{Var}_{cs}(\hat{\beta}) = \sigma_\beta^2 + \overline{\text{se}^2}$, so the empirical-Bayes prior is $\sigma_\beta^2 = \text{Var}_{cs}(\hat{\beta}) - \overline{\text{se}^2}$ (posterior derivation: Seminar 3, problem A3)
- XLK, monthly 1999–2012, 9 sectors: $\hat{\beta} = 1.415$ (s.e. 0.139), $\bar{\beta} = 0.949$, $\sigma_\beta^2 = 0.116 - 0.008 = 0.108$, $w = 0.849$
  - $\beta^V = 0.849 \times 1.415 + 0.151 \times 0.949 = 1.344$; Blume 1.278; beta estimated on 2013–2026: 1.172

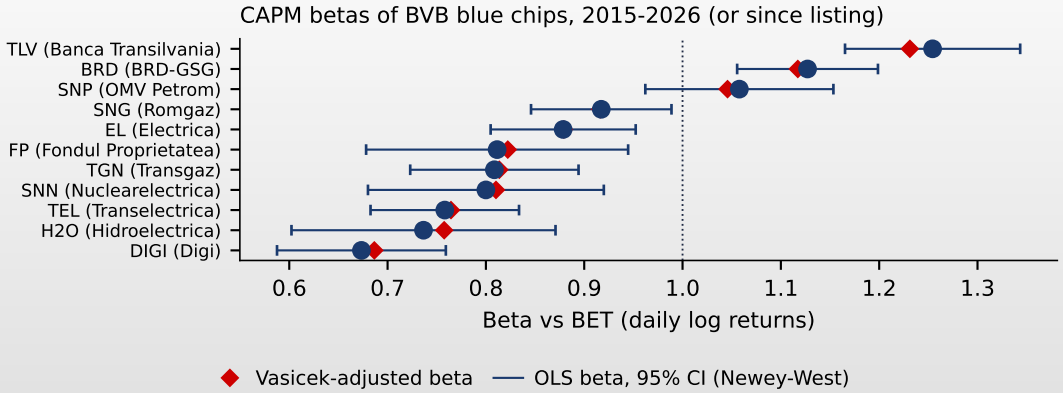
Do Betas Regress Towards One?



- Monthly betas of 9 sector ETFs on SPY: 1999–2012 (horizontal) vs 2013–2026 (vertical)
- Fitted line $\hat{\beta}_{\textbf{later}} = 0.23 + 0.72\,\hat{\beta}_{\textbf{earlier}}$, very close to Blume's $0.33 + 0.67\,\hat{\beta}$
- **RMSE** of the forecast of later betas, $\sqrt{\frac{1}{N}\sum_i (\text{forecast}_i - \hat{\beta}_{i,\textbf{later}})^2}$: raw 0.135, Blume 0.109, Vasicek 0.114
- Shrinkage cuts the forecast error by 15–20%; with 9 sectors the lesson is qualitative

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CAPM Betas on the Bucharest Stock Exchange (1/2)



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CAPM Betas on the Bucharest Stock Exchange (2/2)

- Daily log returns, regressed on BET, from 6 January 2015 or the listing date (Digi, DIGI, 2017; Hidroelectrica, H2O, 2023); days with $|r_t| > 0.5$ removed (one for DIGI, two for Fondul Proprietatea, FP)
- Banks are the most market-sensitive: Banca Transilvania (TLV) 1.25, BRD 1.13; OMV Petrom 1.06; Transelectrica 0.76, Digi 0.67
- R^2: 0.59 for TLV, only 0.21 for DIGI; large index weights raise R^2 mechanically
- Vasicek changes little: the daily estimates are precise (se $\approx$ 0.04–0.07); stale prices bias daily betas of less liquid stocks (Seminar 3, problem B4)

MFM_ch3_bvb_betas

From One Alpha to the GRS Test (1/2)

- **Jensen's alpha** (Jensen (1968)): the market-model intercept α_i
  - The CAPM implies $\alpha_i = 0$ for every asset
- A separate t-test for each asset: XLK has $t = 0.13$
  - False rejections accumulate when the assets are many (Section 8)
- **GRS** (Gibbons, Ross & Shanken (1989)): a joint test of $H_0 : \alpha_1 = \cdots = \alpha_N = 0$

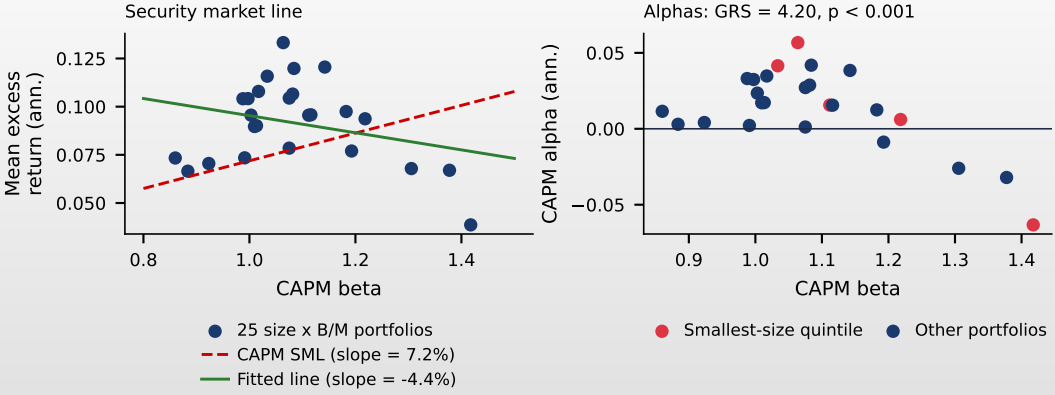
$$W = \frac{T - N - 1}{N} \, \frac{\hat{\boldsymbol{\alpha}}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\alpha}}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \;\sim\; F(N,\, T - N - 1)$$

- Notation
  - N: number of test assets; T: number of months; $\hat{\boldsymbol{\alpha}}$: the N estimated alphas
  - $\hat{\boldsymbol{\Sigma}}$: maximum-likelihood residual covariance (divided by T)
  - $\hat{\mu}_m / \hat{\sigma}_m$: Sharpe ratio of the market

From One Alpha to the GRS Test (2/2)

- $\hat{\boldsymbol{\alpha}}^{\top}\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\alpha}}$: the alphas squared and weighted by the inverse residual covariance
- Decision: reject H_0 when W exceeds the 95% quantile of $F(N, T-N-1)$; F: Fisher distribution
- The distribution is exact, conditional on the factors
  - errors i.i.d. (independent and identically distributed) multivariate Normal, independent of the factors
  - zero mean and constant nonsingular covariance
- Large W: the market is far from the mean–variance frontier spanned by the market and the test assets

The Security Market Line in the Data (1/2)



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The Security Market Line in the Data (2/2)

- Test assets: US stocks sorted each June into 5 size and 5 B/M quintiles, 25 value-weighted portfolios (Fama & French (1993)), July 1963 – July 2026, $T = 757$
- The CAPM line has slope 7.2% a year; the fitted line has slope -4.4%: higher beta, **lower** average return
- Betas span only 0.86–1.42, average excess returns 3.9%–13.3%
- Small-value alpha $+5.7\%$, small-growth -6.3% a year; 10 of 25 alphas have $|t| > 1.96$
- GRS $= 4.20$, p-value 8×10^{-11}: the CAPM is rejected

MFM_ch3_sml

GRS as a Sharpe-Ratio Test

- Identity (Gibbons, Ross & Shanken (1989)): with $\hat{\boldsymbol{\Sigma}}$ and the moments estimated by maximum likelihood

$$\hat{\boldsymbol{\alpha}}^{\top}\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\alpha}} = \widehat{SR}^2(f,\mathrm{R}) - \widehat{SR}^2(f)$$

  - $\widehat{SR}^2(f,\mathrm{R})$: squared Sharpe ratio of the tangency portfolio of the factor and the N test assets; $\widehat{SR}^2(f)$: that of the factor alone
- So GRS measures the Sharpe-ratio gain from the test assets

$$W = \frac{T-N-1}{N}\cdot\frac{\widehat{SR}^2(f,\mathrm{R}) - \widehat{SR}^2(f)}{1+\widehat{SR}^2(f)}$$

- 25 portfolios, monthly Sharpe ratios: market 0.134 (0.47 a year); market plus portfolios 0.405 (1.40 a year)
- $W = 29.24 \times (0.405^2 - 0.134^2)/(1+0.134^2) = 4.20$; the gain expected under H_0 is far smaller (Appendix)

When the GRS Assumptions Fail

- **Heavy tails and volatility clustering** (Chapter 1): the F distribution is no longer exact
  - Residual bootstrap under H_0: resample whole rows of residuals, rebuild returns with $\alpha = 0$ (Seminar 3, problem B7)
  - The bootstrap is robust to non-Normality only if the errors are i.i.d. and independent of the factors
  - GMM Wald test: moments $\varepsilon_t \otimes (1, f_t)$, each residual times 1 and times the factor ($\otimes$: Kronecker product)
  - $W_{\text{GMM}} = \hat{\alpha}^\top \hat{V}_\alpha^{-1} \hat{\alpha} \to \chi^2_N$, $\hat{V}_\alpha$: HAC estimate of $\text{Var}(\hat{\alpha})$; robust to heteroskedasticity and autocorrelation
- **Rank condition**: $\hat{\Sigma}$ has rank at most $T - 2$
  - GRS is not computable when $N \geq T - 1$, and its power falls well before that point
- Tests built for many assets: Appendix

A Flat SML: Black and Betting Against Beta



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia (2009 upload); public domain; Wikimedia Commons

- Black (1972): with restricted borrowing, beta pricing holds with a **zero-beta** rate above R_f in place of the risk-free rate
- Prediction: the SML is **flatter** than the CAPM line and starts at the zero-beta rate
- Frazzini & Pedersen (2014): constrained investors who want more risk buy high-beta assets instead of using leverage; these assets become expensive and earn less
- Their strategy, **BAB** (Betting Against Beta)

$$r_t^{\mathbf{BAB}} = \frac{R_{L,t} - R_{f,t}}{\beta_L} - \frac{R_{H,t} - R_{f,t}}{\beta_H}$$

- L: low-beta portfolio leveraged to beta 1; H: high-beta portfolio de-leveraged to beta 1; BAB has beta zero by construction and a positive average return in their data
- An unscaled low-minus-high-beta spread (Seminar 3, problem C2) is not beta-neutral: its mean mixes the premium with a negative market exposure

What a CAPM Test Establishes

- Roll (1977): the CAPM is a statement about the **true** market portfolio of all wealth; stock indices omit bonds, real estate, private firms and human capital
- Any efficient portfolio prices all assets exactly by beta: with R_f on the CML, with a zero-beta rate (Black (1972)) on the risky-asset frontier (Seminar 3, problem A2)
  - A CAPM test is a joint test of the model and of the efficiency of the proxy
- **Test assets matter**: 9 sector ETFs, January 1999 – July 2026 ($T = 331$): GRS $= 0.53$, $p = 0.85$, not rejected
  - On the same 331 months the 25 size $\times$ B/M portfolios still reject: GRS $= 2.91$, $p = 9 \times 10^{-6}$
  - Sectors mix small and big, value and growth: they do not spread the characteristics that matter
- Failing to reject is weak evidence; evaluate models by how well they price assets with dispersion in expected returns

Arbitrage Pricing Theory

- **APT**: Arbitrage Pricing Theory, Ross (1976)
- Returns follow a K-factor model with small idiosyncratic risks

$$R_{i,t} = \mathbb{E}[R_i] + \boldsymbol{\beta}_i^\top \mathrm{f}_t + \varepsilon_{i,t}, \qquad \mathbb{E}[\mathrm{f}_t] = 0$$

- **No arbitrage** (no riskless profit without investment) implies, approximately

$$\mathbb{E}[R_i] - R_f = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$$

- $\boldsymbol{\beta}_i$: the K loadings of asset i; f_t: the factor surprises, with mean 0; $\boldsymbol{\lambda}$: the **risk premia**, the extra expected return per unit of each exposure
- The theory does not say **which** factors: they must be found in the data

Building Factors: Long–Short Portfolios

size: NYSE median; B/M: 30% / 40% / 30%

	Low (growth)	Neutral	High (value)
Small	S_L	S_N	S_H
Big	B_L	B_N	B_H

$$\text{SMB} = \tfrac{1}{3}(S_L + S_N + S_H) - \tfrac{1}{3}(B_L + B_N + B_H)$$

$$\text{HML} = \tfrac{1}{2}(S_H + B_H) - \tfrac{1}{2}(S_L + B_L)$$

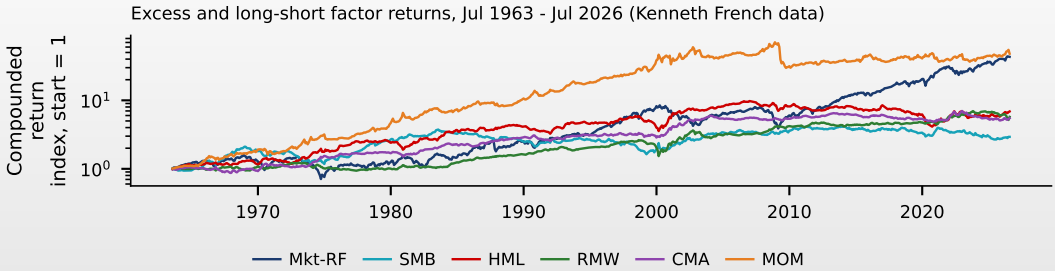
- A **characteristic-sorted factor** buys stocks with a high value of a characteristic and sells those with a low value
- Zero net investment: the return is a pure spread, $f_t = R_{\textbf{high},t} - R_{\textbf{low},t}$
- Fama–French 2×3 sort each June, accounting data of the previous year, value-weighted portfolios
  - **SMB** (Small Minus Big): small minus big stocks
  - **HML** (High Minus Low): high B/M (value) minus low B/M (growth)
- The same construction on profitability and investment gives RMW and CMA (next slide)

The Standard Factor Models

Model	Factors	Reference	Motivation
CAPM	Mkt-RF	Sharpe (1964)	equilibrium
FF3	Mkt-RF, SMB, HML	Fama & French (1993)	size and B/M effects (Fama & French (1992))
Carhart	FF3 + MOM	Carhart (1997)	fund evaluation
q-factor	market, size, investment, ROE	Hou, Xue & Zhang (2015)	firms' investment decisions
FF5	Mkt-RF, SMB, HML, RMW, CMA	Fama & French (2015)	dividend-discount model

- FF3: $R^e_{i,t} = \alpha_i + b_i\,\mathrm{MKT}_t + s_i\,\mathrm{SMB}_t + h_i\,\mathrm{HML}_t + \varepsilon_{i,t}$, with MKT = Mkt-RF and loadings b_i, s_i, h_i on the three factors; a good model leaves time-series alphas close to zero (GRS not rejected)
- Factors beyond FF3 (ROE: Return On Equity)
  - **RMW** (Robust Minus Weak): high minus low operating profitability
  - **CMA** (Conservative Minus Aggressive): low minus high asset growth
  - **MOM** (UMD, Up Minus Down): high minus low return from month $t-12$ to $t-2$; past winners keep winning over 3–12 months (Jegadeesh & Titman (1993))
- FF3 uses its own SMB (2×3 sorts on size and B/M); the FF5 SMB averages size spreads from three sorts, so FF3 is not nested in FF5

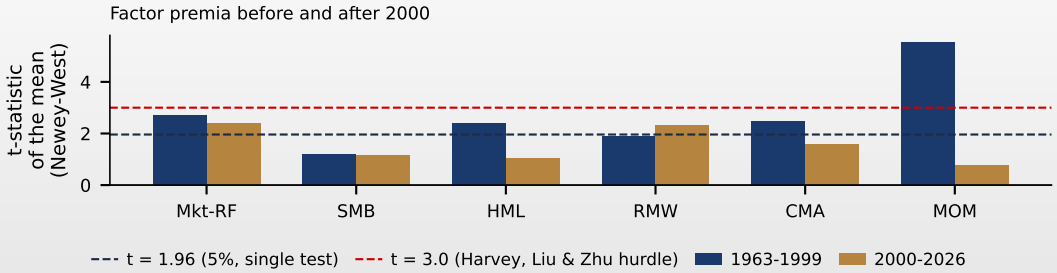
Six Factors over Sixty-Three Years



- Compounded monthly factor returns from 1, log scale: excess or long–short returns, so no interest on capital or collateral; the curves are not the wealth of a funded investment
- July 1963 – July 2026: Mkt-RF 43.2, MOM 48.0, HML 6.9, RMW 5.7, CMA 5.5, SMB 2.9
- Momentum peaked before 2009 and lost heavily in the 2009 rebound: a **momentum crash**
- The value factor was flat or losing from about 2007 to 2020
- Average annual returns: Mkt-RF 7.2%, MOM 7.3%, HML 3.6%, RMW 3.1%, CMA 3.0%, SMB 2.3%

MFM_ch3_factor_cum

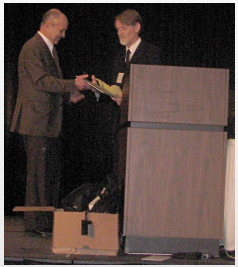
Do the Premia Survive?



- t-statistic of the mean monthly factor return (Newey–West), 1963–1999 vs 2000–2026
- Momentum: $t = 5.53$ before 2000, 0.75 after; mean 10.7% vs 2.5% a year
- Value (HML): $t = 2.39$ then 1.02; mean 4.2% vs 2.8%
- Profitability (RMW) is the exception: $t = 1.90$ then 2.31; the market premium holds in both halves ($t = 2.70$ and 2.39)
- A calendar split is not a publication event: dating the decay needs the design of Section 12

MFM_ch3_factor_tstats

The Two Passes



Eugene Fama, Morgan Stanley–AFA Award, 2008

Photo: TonyTheTiger (2008); CC BY-SA 3.0; Wikimedia Commons

- Fama & MacBeth (1973) estimate the risk premia $\boldsymbol{\lambda}$ of $\mathbb{E}[R_i^e] = \lambda_{\mathbf{0}} + \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$ (Fama: Nobel Prize 2013)
- **Pass 1**: time series, one regression per asset
  - returns $T \times N$, factors $T \times K \Rightarrow$ betas $\hat{\mathrm{B}}$, $N \times K$

$$R_{i,t}^e = a_i + \boldsymbol{\beta}_i^\top \mathrm{f}_t + \varepsilon_{i,t}$$

- **Pass 2**: cross section, each month one regression of the N returns on $\mathrm{X} = [1, \hat{\mathrm{B}}]$

$$R_{i,t}^e = \lambda_{\mathbf{0},t} + \hat{\boldsymbol{\beta}}_i^\top \boldsymbol{\lambda}_t + u_{i,t}, \qquad \hat{\boldsymbol{\lambda}} = \tfrac{1}{T} \textstyle\sum_t \hat{\boldsymbol{\lambda}}_t$$

- $\lambda_{\mathbf{0}}$: pricing error common to all assets, zero if the model is right; a_i: pass-1 intercept; $u_{i,t}$: pass-2 residual; $\hat{\boldsymbol{\lambda}}$: average of the monthly slopes
- The method allows factors that are **not** traded portfolios, and firm characteristics as regressors

Standard Errors for Risk Premia (1/2)

- **Fama–MacBeth**: the monthly $\hat{\lambda}_{k,t}$ form a sample of size T

$$\text{se}_{FM}(\hat{\lambda}_k) = s(\hat{\lambda}_{k,t})/\sqrt{T}$$

  - $s(\cdot)$: the sample standard deviation over the T months
  - robust to cross-sectional correlation of the residuals
  - if $\hat{\lambda}_{k,t}$ is autocorrelated, use Newey–West on this series
- **Misspecification**: the Fama–MacBeth, Newey–West and Shanken (next slide) errors assume a correct model
  - pricing errors $\mathsf{e} = \boldsymbol{\mu} - \mathsf{X}\boldsymbol{\gamma} = 0$; $\boldsymbol{\mu}$: mean excess returns, $\boldsymbol{\gamma} = (\lambda_0, \boldsymbol{\lambda}^\top)^\top$
  - when GRS rejects the model, the noise in $\hat{\mathsf{B}}$ multiplies e and adds variance (Kan, Robotti & Shanken (2013); GMM form in the Appendix)

Standard Errors for Risk Premia (2/2)

- **Errors in variables**: pass 2 uses $\hat{\beta}$, not β, so se_{FM} is too small
- Shanken (1992) scales the non-factor part of the variance by $(1 + c)$

$$\text{Var}_{\text{Sh}}(\hat{\boldsymbol{\lambda}}) = (1 + c)\left[\text{Var}_{FM}(\hat{\boldsymbol{\lambda}}) - \tfrac{1}{T}\boldsymbol{\Sigma}_f\right] + \tfrac{1}{T}\boldsymbol{\Sigma}_f, \qquad c = \hat{\boldsymbol{\lambda}}^\top \boldsymbol{\Sigma}_f^{-1}\hat{\boldsymbol{\lambda}}$$

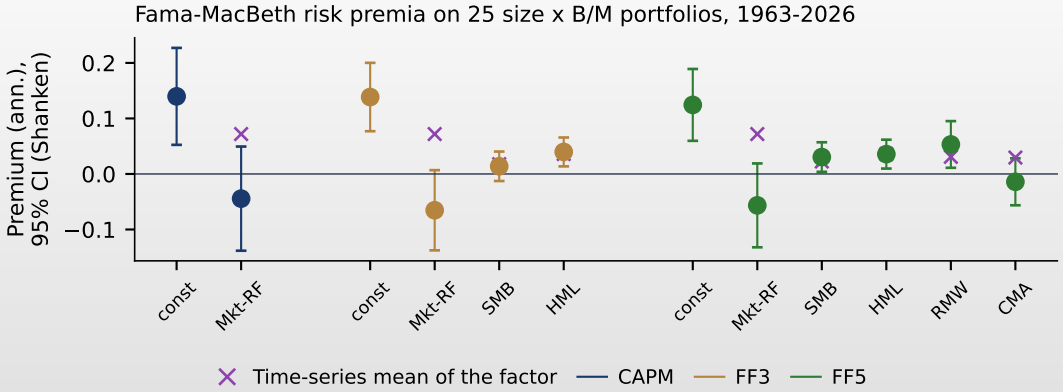
  - Var_{FM}: the Fama–MacBeth variance, $s^2(\hat{\boldsymbol{\lambda}}_t)/T$; $\boldsymbol{\Sigma}_f$: factor covariance
  - c: squared Sharpe ratio of the premia per period, 0.03 for FF3 here, larger for some non-traded factors
- Small c (traded factors, monthly data): the correction is small; large c: Fama–MacBeth t-statistics are overstated

Worked Example: Four Standard Errors for the Value Premium

FF3, HML	$\hat{\lambda}$	**s.e.**	t	**Significant at 5%?**
Fama–MacBeth	3.97%	1.33%	2.99	yes
Newey–West (6 lags)	3.97%	1.62%	2.45	yes
Shanken	3.97%	1.33%	2.99	yes
Kan–Robotti–Shanken	3.97%	1.32%	3.00	yes

- 25 size $\times$ B/M portfolios, 1963–2026; Shanken barely moves the error ($c = 0.03$ per month)
- Autocorrelation of the monthly slopes $\hat{\lambda}_t$ matters most here: Newey–West, 6 lags by the rule for $T = 757$
- FF5: for RMW, the misspecification-robust s.e. rises from 2.1% to 2.6%, and t falls from 2.5 to 2.1
- At the $t > 3$ hurdle of Section 8, HML is borderline under every choice (2.45–3.00)

Estimated Risk Premia (1/2)



MFM_ch3_fama_macbeth

Estimated Risk Premia (2/2)

- 25 size $\times$ B/M portfolios, 1963–2026; bars: 95% intervals with Shanken errors; crosses: factor means
- CAPM: intercept 14.0% (s.e. 4.4%), market premium -4.4% (s.e. 4.8%)
- FF3: SMB 1.4% (1.4%), HML 4.0% (1.3%); FF5: RMW 5.3% (2.1%), CMA -1.4% (2.2%)
- For a traded factor, λ should equal its time-series mean; HML 4.0% vs 3.6%, close; market (FF3) -6.5% vs 7.2%, far apart
- Market betas span only 0.86–1.42 in the CAPM (cross-sectional standard deviation 0.14) and 0.89–1.14 in FF3 (0.07): λ_m is **weakly identified**, and a large intercept absorbs the level (Lewellen, Nagel & Shanken (2010))

MFM_ch3_fama_macbeth

Useless and Weak Factors

- A **useless factor** g_t is independent of all returns: $\beta_g = 0$, so $\hat{\beta}_{i,g} = O_p(T^{-1/2})$, that is, of the order of $1/\sqrt{T}$: pure noise
- Pass 2 divides by this noise: $\hat{\lambda}_g$ does not converge to zero and its t-test over-rejects when the model is misspecified (Kan & Zhang (1999))
- Simulation on the 25 portfolios, CAPM plus an independent Normal factor, 2000 replications, nominal level 5%
  - $\lambda_g = 0$ rejected in 60% (Fama–MacBeth errors) and 27% (Shanken errors) of the replications
  - First-pass Wald test of $\beta_{i,g} = 0$ for all i: rejected in 6.7%; run it before pass 2
- **Weak identification**: betas close to zero (a weak factor) or nearly constant across assets (the market here)
  - identification-robust tests (Kleibergen (2009))
  - with many assets, consistency of pass 2 depends on factor strength (Anatolyev & Mikusheva (2022))

A Skeptical Appraisal: Adding 30 Industry Portfolios

	OLS R^2		GLS R^2	
Model	25	25 + 30	25	25 + 30
CAPM	0.08	0.00	0.06	0.01
FF3	0.66	0.22	0.21	0.08
FF5	0.76	0.24	0.26	0.10

Monthly 1963–2026, free intercept and premia

- The 25 portfolios have a strong three-factor structure: any factors correlated with SMB and HML fit their means, so a high OLS R^2 is a low hurdle (Lewellen, Nagel & Shanken (2010))
- Three checks
  - Broaden the test assets: FF3 falls from 0.66 to 0.22 once 30 industries are added
  - Report the **GLS** (Generalised Least Squares) R^2, weighted like GRS by $\hat{V}^{-1}$, the inverse covariance matrix of the returns: never above 0.26
  - Impose $\lambda = \mathbb{E}[f]$ for traded factors, no intercept: FF3 on the 55 assets gives $R^2 = -0.27$
- Stocks instead of portfolios: more dispersion in betas, larger errors in variables (Ang, Liu & Schwarz (2020))

Factors as a Stochastic Discount Factor



Lars Peter Hansen, Nobel Prize in Economics 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

- **SDF** (Stochastic Discount Factor) m_t: prices all excess returns, $\mathbb{E}[m_t R^e_{i,t}] = 0$
- A linear factor model is an SDF, $m_t = 1 - \mathrm{b}^\top(\mathrm{f}_t - \mathbb{E}\mathrm{f})$; b: the SDF loadings, $\mathbb{E}\mathrm{f}$: the factor means
- Substitute: $\mathbb{E}[R^e_i] = \mathrm{Cov}(R^e_i, \mathrm{f}_t)^\top \mathrm{b} = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$ with $\boldsymbol{\lambda} = \boldsymbol{\Sigma}_f \mathrm{b}$: the beta pricing model of Section 6
- $\mathrm{b} \neq 0$ asks whether a factor helps price the assets; $\boldsymbol{\lambda} \neq 0$ whether it is rewarded
- These moment conditions are estimated by GMM, Hansen (1982); survey of the discount-rate view: Cochrane (2011)

Comparing Models: Maximum Sharpe Ratios

Max. Sharpe ratio (annual)	CAPM	FF3	Carhart	FF5	FF5 + MOM
1963–2026 ($T = 757$)	0.47	0.66	0.97	0.97	1.16
2000–2026 ($T = 319$)	0.48	0.54	0.67	0.97	1.02

- Barillas & Shanken (2018): judge a model by $SR^2(\mathrm{f}) = \boldsymbol{\mu}_f^\top \boldsymbol{\Sigma}_f^{-1} \boldsymbol{\mu}_f$ of its factors ($\boldsymbol{\mu}_f$, $\boldsymbol{\Sigma}_f$: means and covariance of the factors), the squared Sharpe ratio of their tangency portfolio; the test assets do not matter for the comparison
- Nested models: in-sample SR^2 can only rise, so test the alpha from the regression of the added factor on the others (spanning regression)
  - HML on the other five factors: $\alpha = 0.9\%$ a year ($t = 0.75$); since 2000 -1.7% ($t = -0.81$): no significant incremental contribution of HML, as in Fama & French (2015)
- Non-nested FF5 vs Carhart: difference in annual SR^2 0.00, 95% interval $[-0.57, 0.66]$; since 2000 0.50, $[-0.24, 1.48]$ (stationary bootstrap, Politis & Romano (1994))
- Another model-independent metric, the Hansen–Jagannathan distance: Appendix

The Factor Zoo

- Cochrane (2011) called the growing list of new factors a "zoo"
- Harvey, Liu & Zhu (2016) catalogue hundreds of published factors (316 in their list)
- Each was tested as if it were the only one: $|t| > 1.96$ at 5%
- With many tests, some pass by chance: **data mining**, also called p-hacking
- Their proposal: a new factor needs $|t| > 3.0$

Controlling False Discoveries

- **FDR** (False Discovery Rate): expected share of false rejections among all rejections
- **FWER** (Family-Wise Error Rate): probability of at least one false rejection (Chapter 2)
- Notation of the table: M: number of tests; α: significance level (not a pricing error); $p_{(1)} \leq \cdots \leq p_{(M)}$: sorted p-values; $H_{(j)}$: the hypothesis with the j-th smallest; **BH**: the Benjamini–Hochberg procedure

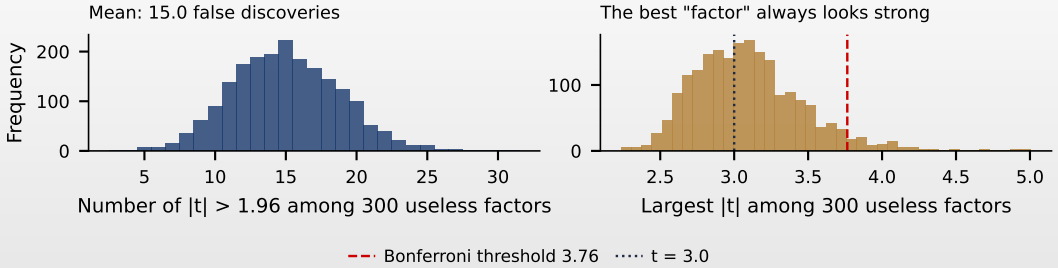
Procedure	**Controls**	**Reject $H_{(j)}$ if**	**Valid under**
Bonferroni	FWER	$p_{(j)} \leq \alpha/M$	any dependence
Holm (Holm (1979))	FWER	$p_{(k)} \leq \alpha/(M-k+1)$ for all $k \leq j$	any dependence
BH (Benjamini & Hochberg (1995))	FDR	$j \leq$ largest k with $p_{(k)} \leq k\alpha/M$	independence, positive dep.
BY (Benjamini & Yekutieli (2001))	FDR	as BH, thresholds divided by $\sum_{k=1}^{M} 1/k$	any dependence

- Positive dependence: the test statistics of the true nulls move up together (PRDS condition, Benjamini & Yekutieli (2001))
- FDR control suits a search over many candidate factors

Worked Example: Ten Candidate Factors

- Sorted p-values: 0.0004, 0.002, 0.006, 0.011, 0.018, 0.027, 0.041, 0.070, 0.15, 0.62; significance level $\alpha = 5\%$
- Naive ($p \leq 0.05$): **7** discoveries
- Bonferroni ($p \leq 0.005$): **2**
- Holm: $0.0004 \leq 0.005$, $0.002 \leq 0.0056$, $0.006 \leq 0.00625$, $0.011 > 0.0071$ stop: **3**
- BH thresholds $0.005j$: $p_{(6)} = 0.027 \leq 0.030$ is the last success: **6**
- Seminar 3, problem A1 repeats the procedure on a different vector, with a threshold chart

Simulation: 300 Useless Factors



- 2000 replications; each draws 300 independent t-statistics from the standard Normal distribution (no true premium)
- On average 15.0 useless factors pass $|t| > 1.96$; expected value $300 \times 0.05 = 15$
- On average 0.83 pass $|t| > 3$; at least one passes in 57% of the replications; median largest $|t|$: 3.06
- Bonferroni threshold for 300 tests: $|t| > 3.76$; hurdles rise with the number of tests actually tried, including unpublished ones

MFM_ch3_false_discoveries

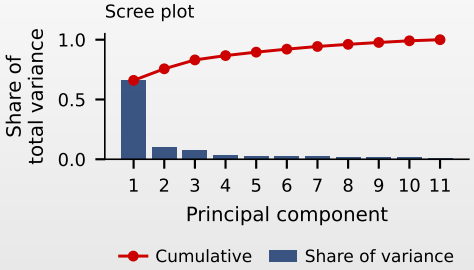
Principal Components in Brief

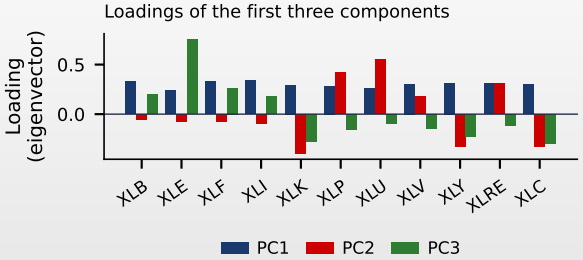
- **Latent factors**: estimated from returns alone, without naming them in advance
- **PCA** (Principal Component Analysis) of the correlation matrix C of standardised returns

$$\mathsf{C} = \mathbf{\Gamma}\mathsf{D}\mathbf{\Gamma}^\top, \qquad \mathsf{y}_t = \mathbf{\Gamma}^\top \mathsf{z}_t$$

  - z_t: standardised returns of day t; y_t: the components
  - $\mathbf{\Gamma}$: eigenvectors in columns; D: diagonal matrix of the eigenvalues ℓ_k
- Eigenvalues $\ell_1 \geq \cdots \geq \ell_N$ (ℓ, not the risk premium λ); eigenvectors are the **loadings**; component k explains ℓ_k/N of the variance
- Two assets with correlation ρ: $\ell = 1 \pm \rho$, eigenvectors $(1,1)/\sqrt{2}$ (the "market") and $(1,-1)/\sqrt{2}$ (the spread); in a crisis ρ rises and PC1 dominates
- Textbook: Härdle & Simar (2019), chapter on principal components; N equicorrelated assets: Seminar 3, problem A8

PCA of the US Sector ETFs





- 11 sector ETFs, daily log returns, 20 June 2018 – 18 September 2026 ($T = 2073$ common days)
- PC1 explains 66.0%; all loadings positive (0.24–0.34); correlation with SPY 0.954: strongly market-related
- PC2 (9.7%): utilities $+0.55$, staples $+0.42$, real estate $+0.31$ vs technology -0.40, consumer discretionary -0.33: **defensive versus growth**, with no name given in advance
- PC3 (7.4%) is dominated by energy; three components explain 83.1%

MFM_ch3_pca

How Many Factors?

- Approximate factor model $x_{it} = \boldsymbol{\Lambda}_i^\top \mathrm{F}_t + e_{it}$, $N, T \to \infty$
  - x_{it}: standardised return; F_t: factors; $\boldsymbol{\Lambda}_i$: loadings; e_{it}: idiosyncratic part
  - $V(k)$: mean squared residual after the first k components; ℓ_k: the k-th largest eigenvalue
- Bai & Ng (2002): $\hat{k} = \arg\min_k \left[\ln V(k) + k \, \frac{N+T}{NT} \ln \min(N, T) \right]$ (criterion IC_{p2}); Ahn & Horenstein (2013): eigenvalue ratio $\hat{k} = \arg\max_k \ell_k / \ell_{k+1}$, no penalty to choose
- The rule of Kaiser (1960) ($\ell_k > 1$) has no asymptotic justification: the noise eigenvalues of a correlation matrix scatter around 1

Panel	N	T	Kaiser	Bai–Ng	Ahn–Horenstein
11 sector ETFs, daily 2018–2026	11	2073	2	5	1
100 size × B/M portfolios, monthly 1963–2026	95	757	3	4	1

- Standardised returns; maximum k: 5 for sectors, 8 for portfolios; 5 portfolios with missing months are dropped
- Ahn–Horenstein picks the dominant market factor (ratio 14.6); its next peak is at $k = 3$ (ratio 3.5), the FF3 structure
- Bai–Ng on 11 sectors hits the upper bound $k = 5$: the penalty is built for large N, so with small N the answer is fragile (▸ Appendix)

Limits of PCA for Asset Pricing

- Loadings are **constant**: PCA cannot follow a stock whose size or value changes
- Components explain **variance**, not necessarily **average returns**
- Individual stocks enter and exit the sample: a stock panel is unbalanced
- Component signs are arbitrary; rotations preserve principal components only within repeated-eigenvalue eigenspaces, while a latent-factor model is identified only up to rotation
- Risk-premium PCA and three-pass premia keep the PCA backbone but target pricing (▸ Appendix); PCA and factor analysis in finance: Franke, Härdle & Hafner (2019)

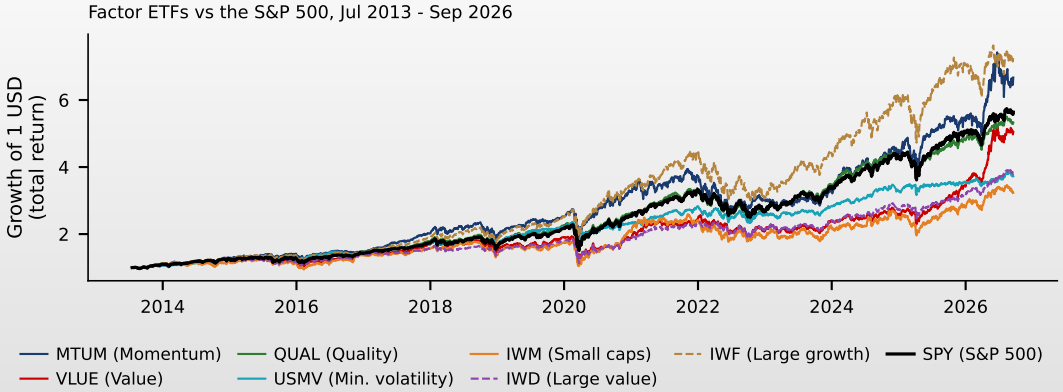
Instrumented PCA

- **IPCA**: Instrumented Principal Component Analysis, Kelly, Pruitt & Su (2019)
- Loadings are functions of L observed characteristics $z_{i,t}$ (size, B/M, momentum, ...)

$$R^e_{i,t+1} = \alpha_{i,t} + \boldsymbol{\beta}_{i,t}^\top \mathbf{f}_{t+1} + \varepsilon_{i,t+1}, \qquad \boldsymbol{\beta}_{i,t}^\top = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\beta$$

- $\boldsymbol{\Gamma}_\beta$ ($L \times K$) and the latent factors f_t are estimated jointly by alternating least squares
- Test: are characteristics priced through betas ("covariances"), or directly as alphas ($\alpha_{i,t} = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\alpha$)?
- Result in their US stock data: a few IPCA factors price the cross section better than observable-factor models with as many factors, and $\boldsymbol{\Gamma}_\alpha$ is insignificant: "characteristics are covariances"; crypto-asset factors: Appendix

Factor ETFs versus the Market (1/2)



MFM_ch3_factor_etfs

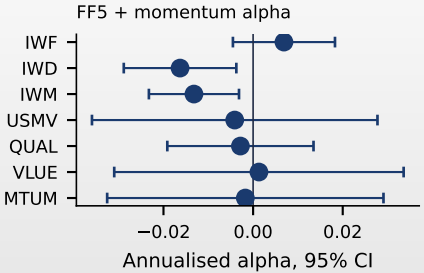
Factor ETFs versus the Market (2/2)

- **Smart beta**: rule-based long-only portfolios that tilt towards factors, sold as ETFs
  - they keep market exposure and need not replicate an academic long–short factor
- Thirteen years of one realisation: not evidence against the premia on its own

- **CAGR** (Compound Annual Growth Rate) of 1 USD (US dollar) invested, dividends re-invested, from 18 July 2013
  - above SPY (14.0%): large growth (IWF) 16.3%, momentum (MTUM) 15.5%
  - below SPY: quality (QUAL) 13.5%, value (VLUE) 13.1%, large value (IWD) 10.7%, minimum volatility (USMV) 10.5%, small caps (IWM, Russell 2000 index) 9.3%

MFM_ch3_factor_etfs

Alphas of Factor ETFs



Factor loadings

	Mkt-RF	SMB	HML	RMW	CMA	MOM
MTUM	1.03	-0.12	0.01	-0.22	-0.05	0.36
VLUE	0.98	0.08	0.34	-0.06	0.09	-0.03
QUAL	0.97	-0.08	-0.04	0.15	0.03	-0.04
USMV	0.74	-0.11	-0.02	0.20	0.21	-0.02
IWM	1.04	0.85	0.08	-0.10	-0.02	0.03
IWD	0.94	-0.04	0.29	0.09	0.19	-0.04
IWF	1.03	-0.11	-0.28	-0.03	-0.12	-0.00

- Daily regressions on FF5 + MOM, Newey–West errors, each ETF on its own trading calendar (2000–2026)
- The loadings match the labels: MTUM on MOM 0.36, VLUE on HML 0.34, IWM on SMB 0.85; IWF −0.28 on HML; USMV market beta 0.74
- Alphas close to zero: MTUM −0.2% ($t = -0.11$), QUAL −0.3%; Russell ETFs IWM −1.3% ($t = -2.57$), IWD −1.6% ($t = -2.54$), consistent with costs and index construction
- R^2 between 0.81 and 0.97: mostly repackaged factor exposure

MFM_ch3_etf_alphas

Timing, Crowding and Costs

- **Factor timing**: switching between factors based on signals (valuation spreads, recent returns)
  - Noisy premia and short histories make timing hard to verify out of sample (Chapter 8)
- **Crowding**: many investors hold the same factor positions
  - Premia can shrink (McLean & Pontiff (2016)), and crowded positions can amplify losses during forced unwinding; the 2009 momentum crash coincided with a sharp rebound of past losers (Daniel & Moskowitz (2016))
- **Implementability**: long–short factors ignore trading costs and short-selling fees; ETFs pay them

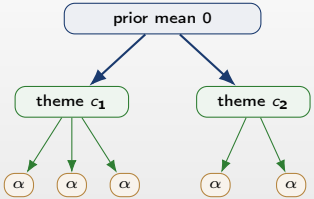
Case Study: Research Questions (1/2)

- **The challenge** to factor research takes two forms (Jensen, Kelly & Pedersen (2023), Introduction)
  - No internal validity: results cannot be reproduced; Hou, Xue & Zhang (2020) replicate only 35% of their anomalies
  - No external validity: results come from testing many factors without controlling the FDR; Harvey, Liu & Zhu (2016) conclude that most claimed findings are likely false
- **Questions**
  - do published factors have significant CAPM alphas in the original data?
  - are these alphas significant out of sample and in other countries?
  - do they remain significant once multiple testing is taken into account?

Case Study: Research Questions (2/2)

- **Why it matters**
  - if most factors are false discoveries, multifactor models and factor investing (Sections 5 and 10) rest on noise
- **A second view of the zoo** (Feng, Giglio & Xiu (2020), Appendix)
  - most new factors are redundant given the existing ones
  - Jensen, Kelly & Pedersen instead treat the zoo as correlated evidence on a few economic themes
- **Link to this chapter**
  - Jensen's alpha (Section 4) is the object tested
  - Benjamini–Yekutieli (Section 8) is the frequentist benchmark
  - Bayesian shrinkage extends the Vasicek beta (Section 3) to alphas

Case Study: Data and Method (1/2)



each factor's alpha borrows strength from the other factors of its theme

- **Data** (Section II): 153 factors in 93 countries
  - terciles of each characteristic, capped value weights
  - long the side with the higher expected return in the original paper
  - each factor scaled to a 10% annual idiosyncratic volatility
- **Themes**: 13 clusters of US factors
  - hierarchical clustering on the correlation of CAPM residuals
- **Model** (Section I): $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$
  - $f_{i,t}$: return of factor i; r_t^m: market excess return
  - alphas $\alpha_i = c_j + s_n + w_i$: theme, signal and factor components, prior mean 0

Case Study: Data and Method (2/2)

- One factor: the posterior mean shrinks the estimated alpha (eq. 5)

$$\mathbb{E}(\alpha \mid \hat{\alpha}) = \kappa \hat{\alpha}, \qquad \kappa = \frac{1}{1 + \sigma^2/(\tau^2 T)}$$

  - σ: residual volatility; τ: dispersion of the true alphas; T: number of months
  - short, noisy samples are shrunk more, as in Vasicek (Section 3)
- **Replication rate**: share of the factors significant in the original studies whose alpha is significant in the new data
  - Bayesian discovery: posterior mean / posterior s.d. ≥ 1.96
- Full hierarchy and estimation: ▸ Appendix

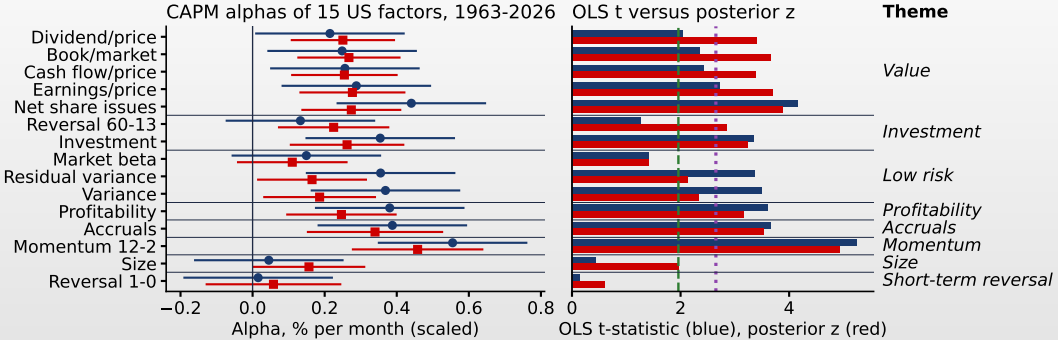
Case Study: Key Results

Replication rates, US (Figure 1)

Hou, Xue & Zhang (2020), raw returns	35.0%
Jensen, Kelly & Pedersen, raw returns	55.6%
Raw returns, 119 originally significant	61.3%
CAPM alpha, OLS	82.4%
CAPM alpha, Benjamini–Yekutieli	75.6%
CAPM alpha, empirical Bayes	82.4%
World, empirical Bayes	82.4%

- Gap with Hou, Xue & Zhang (2020): testing alphas instead of raw returns is the largest step (61.3% → 82.4%); the sample and the factor construction add almost as much (35.0% → 55.6%)
- Bayesian FDR 0.1%: shrinkage controls false discoveries without the power loss of a higher t hurdle
- **Decay** after the original sample
  - mean alpha 0.49% a month in sample, 0.26% afterwards (−47%)
  - slope of post-sample on in-sample alphas 0.26 (Figure 8), against about 0.9 from shrinkage alone
- The strongest factors decay more: arbitrage or data mining (Section 12)
- Complete results: ▸ Appendix

Case Study: Replication on Course Data (1/2)



MFM_ch3_jkp_replication

Case Study: Replication on Course Data (2/2)

- 15 US long–short factors from value-weighted sorts of the Kenneth French Data Library, July 1963 – July 2026 ($T = 757$); CAPM alphas scaled to 10% idiosyncratic volatility; themes of Jensen, Kelly & Pedersen (authors' cluster labels)
- Replication rates: OLS 73.3% (11 of 15), Benjamini–Yekutieli 53.3% (8; critical $|t| = 2.65$), empirical Bayes 86.7% (13); details: Appendix
- Shrinkage towards the theme, not only towards zero: long-term reversal (investment theme) 0.13% $\rightarrow$ 0.23% a month ($t = 1.25$, posterior $z = 2.85$)
- Failures: market beta and short-term reversal ($z \leq 1.41$); size only just passes ($z = 1.96$), echoing the paper, where size is one of the two themes below 50%

MFM_ch3_jkp_replication

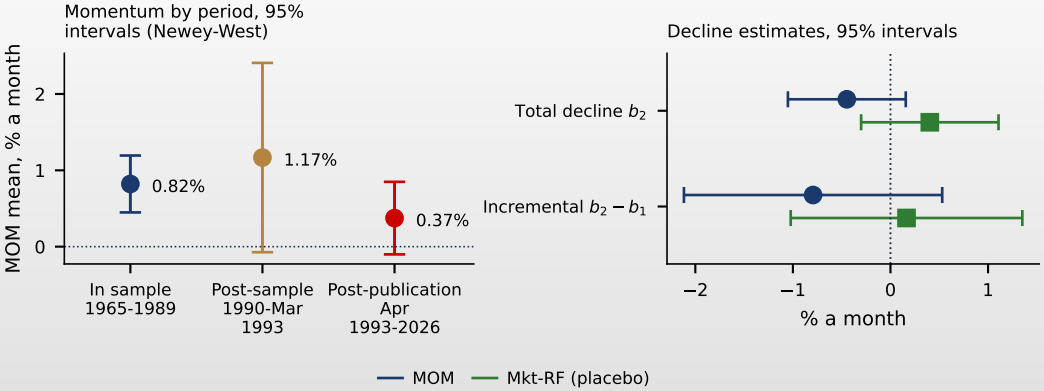
Case Study: Limits and Open Questions

- **Limits of the course version**
  - 15 US signals instead of 153 in 93 countries
  - sort breakpoints of the Kenneth French Data Library instead of non-micro terciles with capped weights
  - one region, so no signal level: $\tau_s = 0$ (τ_s: dispersion of the signal components s_n)
  - the rate depends on the theme structure: with a single common theme the empirical-Bayes rate is 100%
- **Limits stated by the authors** (Appendix)
  - CAPM as the only benchmark; uniform construction of the characteristics; unobserved factors in the prior
- **Open questions**: the paper's data end in 2020
  - did the posterior alphas and the replication rate hold in 2021–2026?
  - can a prior estimated on US factors sharpen short-sample BVB factors?
- Data and code of the paper: jkpfactors.com; course project: Seminar 3, C3

Open Questions and the Discovery Loop

- **Open questions**
  - why do factor premia decay after publication: arbitrage, data mining or a change in risk?
  - does the decay differ across factors and markets?
- Evidence so far
  - McLean & Pontiff (2016): across 97 predictors, returns are 26% lower out of sample and 58% lower after publication; the gap is attributed to publication-informed trading
  - Jensen, Kelly & Pedersen (2023): most factors replicate, but the strongest decay most; Chen & Zimmermann (2022): open data and code for the published predictors
- **One pass of the loop**
  - The assistant drafts: a literature table with DOIs (digital object identifiers), competing hypotheses with their predicted patterns, the regression code, a robustness plan
  - The human fixes before running: the event dates (Jegadeesh & Titman (1993): sample ends 1989, published March 1993), the factor set, the multiple-testing correction, the placebo
  - Checks: every DOI resolves; the chapter's split-sample t-statistics are reproduced first; the data vintage is recorded
- Reusable prompts and the full checklist: ▸ Appendix

Mini-Case: Was Momentum Arbitraged Away? (1/2)



MFM_ch3_ai_discovery

Mini-Case: Was Momentum Arbitraged Away? (2/2)

- Design of McLean & Pontiff (2016): $\text{MOM}_t = a + b_1\text{PostSample}_t + b_2\text{PostPub}_t + e_t$, monthly, January 1965 – August 2026 ($T = 740$), Newey–West errors; post-sample: January 1990 – March 1993; post-publication: from April 1993
- PostSample_t, PostPub_t: dummies equal to 1 in those periods; a: the in-sample mean; b_2: total decline relative to the in-sample mean; $b_2 - b_1$: incremental effect of publication
- $a = 0.82\%$ a month ($t = 4.32$); $b_2 = -0.45\%$ ($t = -1.45$, interval $[-1.05\%, 0.16\%]$): 54% lower; $b_2 - b_1 = -0.79\%$ ($t = -1.17$); placebo Mkt-RF $b_2 = +0.40\%$ ($t = 1.12$)
- Close to the 58% of McLean & Pontiff (2016), but one factor cannot reject $H_0 : b_2 = 0$ at 5%; arbitrage is one interpretation, not an identified cause

MFM_ch3_ai_discovery

Your Turn: A Project Seed

- **Questions**
  - do the Fama–French factors and momentum decay after publication?
  - does the decay differ across factors?
- **Design**
  - pooled regression of SMB, HML, RMW, CMA and MOM on factor-specific post-sample and post-publication dummies
  - Mkt-RF as placebo; errors robust to cross-factor correlation and autocorrelation
  - testing for a larger decay where arbitrage is easier needs stock-level proxies (size, liquidity, short-sale costs)
- **Replicate first**: the split-sample t-statistics and the mini-case estimates
- **Deliverables**
  - a pre-registration note (H_0, H_1, event dates from each paper, factor set, corrections), committed before the results
  - code as a Quantlet; a report with Newey–West inference, placebo periods and power
- **Logs**: prompts in `AI_USE.md`; caught errors in `AI_ERRORS.md`

Key Takeaways: An Empirical Workflow

1. **Estimate exposures** with HAC errors, report sample and proxy, shrink extreme betas (Blume, Vasicek)
2. **Test the pricing restriction jointly**: GRS measures the Sharpe-ratio gain from the test assets; the CAPM fails on the 25 size $\times$ B/M portfolios (flat SML, GRS $= 4.20$)
3. **Estimate premia** in two passes with errors robust to estimated betas and misspecification; a high R^2 on 25 portfolios is a low hurdle
4. **Guard against the zoo**: multiple-testing hurdles, spanning tests, model comparison by maximum Sharpe ratios
5. **Replicate**: latent factors and out-of-sample evidence; factor ETFs are cheap factor exposure, not alpha

⊡ Open questions

- why do premia decay after publication?
- how are assets priced when betas vary over time?
- which factors explain expected returns in small markets such as the BVB?

Check Yourself, and Next

Retrieval questions

- Why does the HAC standard error of a beta differ from the classical one, while the beta does not?
- Why can GRS fail to reject the CAPM on sector ETFs but reject it on the 25 portfolios over the same months?
- Why does a useless factor look priced in pass 2?
- Why can the empirical-Bayes replication rate exceed the Benjamini–Yekutieli rate?

Links to other chapters

- Chapter 2: an alpha is an anomaly or a missing risk factor, the joint hypothesis problem
- Chapter 4: factor covariance models and shrinkage for Markowitz
- Chapters 5–6: time-varying volatility, correlation and betas
- Chapter 8: backtesting factor strategies without overfitting
- Chapter 13: machine learning for the cross section of returns

Appendix: GRS under the Null Hypothesis

- Under H_0 and the exact assumptions, $W \sim F(N, T-N-1)$, so $\mathbb{E}[W] = \frac{T-N-1}{T-N-3}$
- Invert the Sharpe-ratio form of W: the squared Sharpe-ratio gain expected under H_0 is

$$\mathbb{E}\left[\widehat{SR}^2(f,\mathrm{R}) - \widehat{SR}^2(f)\right] \approx \frac{N\left(1+\widehat{SR}^2(f)\right)}{T-N-3}$$

- 25 portfolios, $T = 757$: $25 \times 1.018/729 = 0.035$ (monthly), against $0.405^2 - 0.134^2 = 0.146$ observed
- In-sample maximisation over N assets always adds some Sharpe ratio: the test asks whether the gain exceeds this noise level
- 9 sector ETFs, $T = 331$: observed gain $0.191^2 - 0.146^2 = 0.015$, below the expected $9 \times 1.021/319 = 0.029$

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Appendix: Tests with Many Assets

- $\hat{\boldsymbol{\Sigma}} = \hat{\mathbf{E}}^\top \hat{\mathbf{E}}/T$ has rank at most $T - 2$ (two parameters per regression); $\hat{\mathbf{E}}$: the $T \times N$ matrix of residuals
  - Example: 100 size $\times$ B/M portfolios on 8 years of monthly data ($T = 96$): $\hat{\boldsymbol{\Sigma}}$ is singular, GRS does not exist
  - Well before this point the inverse is badly estimated and power falls
- Tests for many assets that avoid $\hat{\boldsymbol{\Sigma}}^{-1}$: standardised sums of squared t-statistics, Pesaran & Yamagata (2024)
- Power enhancement by screening large alphas: Fan, Liao & Yao (2015)

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Appendix: Misspecification-Robust Standard Errors

- Fama–MacBeth and Shanken errors assume a **correct** model: all pricing errors $e = \mu - X\gamma$ are zero ($X = [1, B]$)
- GRS rejects the model here, so γ is a pseudo-true value; the noise in $\hat{B}$ multiplies e and adds a variance term (Kan, Robotti & Shanken (2013))
- Write both passes as one exactly identified GMM system: moments $(R_t - a - Bf_t) \otimes (1, f_t)$ and $X^\top(R_t - X\gamma)$; the sandwich variance with the full Jacobian is the robust one
  - Setting $e = 0$ in the Jacobian gives the correct-model, heteroskedasticity-robust version of Jagannathan & Wang (1998)
- FF5 on the 25 portfolios (RMW: Worked Example slide): s.e. of CMA 2.2% $\to$ 2.7%, intercept 3.3% $\to$ 3.8%; HML changes by less than 0.05 percentage points
- Advanced course: Generalized Method of Moments

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Appendix: Hansen–Jagannathan Distance and Bayesian Selection

- **Hansen–Jagannathan distance** (Hansen & Jagannathan (1997)): least-squares distance of the model SDF from the set of valid SDFs

$$\delta^2 = \min_{\mathsf{b}} \; \mathrm{e}(\mathsf{b})^\top \, \mathbb{E}[\mathrm{R}^e \mathrm{R}^{e\top}]^{-1} \, \mathrm{e}(\mathsf{b}), \qquad \mathrm{e}(\mathsf{b}) = \mathbb{E}[m_t(\mathsf{b}) \, \mathrm{R}_t^e]$$

  - Excess returns R_t^e with the normalisation $\mathbb{E}[m] = 1$; for gross returns R_t: $\mathrm{e} = \mathbb{E}[m\mathrm{R}] - 1$, and the SDF needs a free scale ($\mathbb{E}[m] = 1/(1 + R_f)$)
  - Same weighting matrix for every model, so distances are comparable; the test statistic has a weighted χ^2 limit
- Zoo-scale selection: Bayesian averaging over about 2×10^{15} (two quadrillion) factor models, robust to weak factors (Bryzgalova, Huang & Julliard (2023))
- SDF shrinkage on characteristic-based factors: a few principal components of anomaly returns capture the SDF (Kozak, Nagel & Santosh (2020))
- Common lesson: compare models by a single, model-independent metric, with its sampling uncertainty

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Appendix: Taming the Zoo with Double-Selection LASSO

- Feng, Giglio & Xiu (2020): does a new factor add information given the existing zoo?
- **Double-selection LASSO** (Least Absolute Shrinkage and Selection Operator)
  - Step 1: a LASSO of average test-asset returns on their covariances with hundreds of factors selects the controls that price the cross section
  - Step 2: a second LASSO selects the factors whose covariances explain those of the new factor, so that omitted controls do not bias its SDF coefficient
  - Step 3: the SDF coefficient of the new factor is estimated with the union of both sets of controls and tested
- Result: most new factors are redundant; only a few add pricing information
- The case study reaches a compatible conclusion: the zoo is highly redundant, and a few themes carry the information

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Appendix: More on the Number of Factors

- Onatski (2010): separates factor eigenvalues from the edge of the noise eigenvalue distribution
- Kaiser ($\ell_k > 1$) on a correlation matrix: with N assets and pure noise the eigenvalues spread around 1 with a width that depends on N/T, so the rule has no asymptotic justification
- Report the range of estimates and the sensitivity to $k_{\max}$, not a single number (Seminar 3, problem B3)

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Appendix: Risk-Premium PCA and Three-Pass Premia

- **RP-PCA** (Risk-Premium PCA, Lettau & Pelger (2020)): principal components of $\frac{1}{T}\mathbf{X}^\top\mathbf{X} + \gamma\bar{\mathbf{x}}\bar{\mathbf{x}}^\top$
  - $\mathbf{X}$: the $T \times N$ matrix of excess returns, $\bar{\mathbf{x}}$: their means, γ: the weight on average returns
  - $\gamma = -1$ gives covariance PCA; $\gamma > -1$ also rewards factors that explain average returns
  - In their data it finds factors with higher Sharpe ratios and smaller pricing errors than PCA
- **Three-pass estimator** (Giglio & Xiu (2021)): the premium of an observed factor g_t when the model omits factors
  - (1) PCA of test-asset returns; (2) cross-sectional regression of means on the latent betas; (3) regress g_t on the latent factors and map their premia to g_t
- Both keep PCA's statistical backbone but target pricing, not only variance; IPCA adds characteristics

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Appendix: IPCA and Crypto-Asset Factors

- IPCA (Kelly, Pruitt & Su (2019)): characteristics predict returns because they measure risk exposure
- IPCA links PCA (statistics) with characteristic sorts (Fama–French): time-varying betas, few factors
- Crypto-assets: market, size and momentum factors price the cross section of coins (Liu, Tsyvinski & Wu (2022)); dimension reduction on crypto and traditional return features: Pele et al. (2023), Jheng et al. (2026)
- Advanced course: Multivariate Statistical Analysis

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Appendix: Case Study, Data and Construction (1/2)

- **Data** (Section II): 153 factors in 93 countries
  - 138 from the list of Hou, Xue & Zhang (2020), plus 15 from the wider literature
  - US returns from 1926, most developed markets from 1986
  - all in US dollars, in excess of the US Treasury bill
- **Construction**, each country and month
  - terciles of the characteristic, breakpoints from non-micro stocks
  - "capped" value weights: market equity winsorised at the NYSE 80th percentile
  - one-month holding period
  - at least 5 stocks per leg and 60 monthly observations
- **Themes**: 13 clusters of US factors (Section II.B)
  - hierarchical clustering, distance 1− correlation of CAPM residuals, Ward linkage

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Appendix: Case Study, Data and Construction (2/2)

- **Model** (Section I): $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$
  - alphas $\alpha_i = c_j + s_n + w_i$ (theme j, signal n, factor i; eq. 21), prior mean 0
  - prior variance $\mathbf{\Omega} = \mathrm{M}\mathrm{M}^\top \tau_c^2 + \mathrm{Z}\mathrm{Z}^\top \tau_s^2 + \mathrm{I}\tau_w^2$ (eq. 23); posterior mean $(\mathbf{\Omega}^{-1} + T\mathbf{\Sigma}^{-1})^{-1} T\mathbf{\Sigma}^{-1}\hat{\boldsymbol{\alpha}}$ (Proposition 4)
  - M, Z: 0/1 matrices linking each factor to its theme and signal; $\tau_c^2, \tau_s^2, \tau_w^2$: variances of the three alpha components; $\mathbf{\Sigma}$: residual covariance
  - empirical Bayes: τ_c, τ_s, τ_w by maximum likelihood of $\hat{\boldsymbol{\alpha}} \sim N(0, \mathbf{\Omega} + \hat{\mathbf{\Sigma}}/T)$ (Section I.C.2)
- **Replication rate**
  - share of the factors significant in the original studies whose alpha is significant in the new data

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Appendix: Case Study, Complete Results

- Raw returns, restricted to the 119 factors significant in the original studies: 61.3% (Figure 1)
- OLS: 98 of 119 US factors; Benjamini–Yekutieli needs $|t| > 2.7$ (Figure 4)
- Bayesian FDR 0.1%, 95% interval $[0, 1\%]$; expected share of true factors 94% (Section III.C)
- Mega and large stocks: 77.3% and 79.8% (Figure 5); 11 of 13 themes at least 50%, all but low leverage and size: not an artefact of micro-caps
- Tangency Sharpe ratio (monthly), US 1972–2020: market 0.13, all factors 0.80 (Figure 14)

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Appendix: Case Study, the Course Replication in Detail

- Benjamini–Yekutieli is the strictest, as in the US rates of the paper (Figure 1)
- Net share issues move towards their theme: 0.44% $\rightarrow$ 0.27% a month
- Posterior s.d. 0.07%–0.10% against OLS s.e. 0.11%: each theme pools the evidence of its members
- **Prior dispersion**: $\hat{\tau}_c = 0.28\%$, $\hat{\tau}_w = 0.04\%$ a month
  - against 0.35% and 0.21% in Table I; 15 signals leave little factor-specific dispersion
- **Themes**: the 15 signals fall into 8 of the paper's 13 themes
  - 5 in value, 3 in low risk, 2 in investment

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Appendix: Case Study, Limits Stated by the Authors and Extensions

- **Limits stated by the authors**
  - CAPM is the only benchmark: low-leverage factors fail on CAPM alpha but matter jointly with other themes (Section III.A)
  - Characteristics are built consistently, sometimes unlike the original paper (Section II.C)
  - Unobserved factors bias the prior: with the scenarios of Harvey, Liu & Zhu (2016) the US rate falls to 81.5% and 79.8% (Figure 10)
- **Complete the course replication** (Seminar 3, C3)
  - justify each sign from the original paper
  - compare the dendrogram with the paper's themes
  - add the Bayesian FDR and the post-sample slope (Figure 8)
- Data and code of the paper: jkpfactors.com, github.com/bkelly-lab/ReplicationCrisis

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Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature**
  - `Summarise the evidence since 2016 on post-publication decay and replication of anomaly returns. For each paper: DOI, sample, number of factors, headline estimate, identification of the decay. Flag entries you are unsure of.`
  - Resolve every DOI on Crossref; check each headline number against the abstract
- **Hypothesis**
  - `Contrast data mining, publication-informed arbitrage and time-varying risk as explanations of decay. For each, describe the predicted pattern across the in-sample, post-sample and post-publication periods, and across high vs low limits-to-arbitrage stocks.`
- **Data and code**
  - `Load the monthly momentum factor from the Kenneth French Data Library; from 1965, regress MOM on a post-sample dummy (1990-01 to 1993-03) and a post-publication dummy (from 1993-04), with Newey-West errors.`
  - Replicate a known number first: momentum $t = 5.53$ before 2000 and 0.75 after (split-sample chart)

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Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**
  - `Write a pre-registered robustness plan: one publication date per factor, a Holm correction across factors, excluding the 2009 momentum crash, the working-paper date instead of the journal date, and 1927-1962 as a placebo period for Mkt-RF, SMB, HML and MOM (RMW and CMA start in July 1963).`
- **Critique**
  - `Referee this: one factor, one publication date, t = -1.45. What is the power to detect a 58% decay? Why does pooling many anomalies change it?`
- **Report**
  - In-sample mean, both dummies with Newey–West errors, the placebo, the data vintage (download date)
  - Pooled factors: errors robust to cross-factor correlation within a month and to autocorrelation
  - Prompts that shaped a decision go into `AI_USE.md`
- The assistant enumerates specifications; the human fixes which ones count before running them

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Appendix: What the Researcher Must Check

- **Wrong event date**: sample end, working paper and journal issue are different dates
  - test: read them from the paper itself; Jegadeesh & Titman (1993): sample ends 1989, published March 1993
- **Choosing the factor after seeing it decay**: fix the set of factors in advance; Holm or BH across them
- **Serially correlated, crash-prone returns**: Newey–West errors; rerun without 2009
- **Data vintages**: the factor library is revised retroactively
  - test: record the download date; reproduce the chapter's split-sample t-statistics
- **Misquoted literature**: a common slip gives 58% as the out-of-sample decline; it is post-publication (26% out of sample)
- **Low power**: with a single factor, H_0 is rarely rejected; report the confidence interval of the decay, not only its sign

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Appendix: Further Study and Code

Advanced courses on quantinar.com

- Generalized Method of Moments
- Multivariate Statistical Analysis

Code and data

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_03
- Factor data: Kenneth French Data Library

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References (1/6)

- Ahn, S.C., Horenstein, A.R. (2013). Eigenvalue ratio test for the number of factors. *Econometrica*, 81(3), 1203–1227.
- Anatolyev, S., Mikusheva, A. (2022). Factor models with many assets: strong factors, weak factors, and the two-pass procedure. *Journal of Econometrics*, 229(1), 103–126.
- Ang, A., Liu, J., Schwarz, K. (2020). Using stocks or portfolios in tests of factor models. *Journal of Financial and Quantitative Analysis*, 55(3), 709–750.
- Bai, J., Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1), 191–221.
- Barillas, F., Shanken, J. (2018). Comparing asset pricing models. *Journal of Finance*, 73(2), 715–754.
- Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- Benjamini, Y., Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics*, 29(4), 1165–1188.
- Black, F. (1972). Capital market equilibrium with restricted borrowing. *Journal of Business*, 45(3), 444–455.
- Blume, M.E. (1971). On the assessment of risk. *Journal of Finance*, 26(1), 1–10.
- Bryzgalova, S., Huang, J., Julliard, C. (2023). Bayesian solutions for the factor zoo: we just ran two quadrillion models. *Journal of Finance*, 78(1), 487–557.

References (2/6)

- Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). The Econometrics of Financial Markets. Princeton University Press.
- Carhart, M.M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52(1), 57–82.
- Chen, A.Y., Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 11(2), 207–264.
- Cochrane, J.H. (2011). Presidential address: discount rates. *Journal of Finance*, 66(4), 1047–1108.
- Daniel, K., Moskowitz, T.J. (2016). Momentum crashes. *Journal of Financial Economics*, 122(2), 221–247.
- Fama, E.F., French, K.R. (1992). The cross-section of expected stock returns. *Journal of Finance*, 47(2), 427–465.
- Fama, E.F., French, K.R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
- Fama, E.F., French, K.R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- Fama, E.F., MacBeth, J.D. (1973). Risk, return, and equilibrium: empirical tests. *Journal of Political Economy*, 81(3), 607–636.
- Fan, J., Liao, Y., Yao, J. (2015). Power enhancement in high-dimensional cross-sectional tests. *Econometrica*, 83(4), 1497–1541.

References (3/6)

- Feng, G., Giglio, S., Xiu, D. (2020). Taming the factor zoo: a test of new factors. *Journal of Finance*, 75(3), 1327–1370.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Frazzini, A., Pedersen, L.H. (2014). Betting against beta. *Journal of Financial Economics*, 111(1), 1–25.
- Gibbons, M.R., Ross, S.A., Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5), 1121–1152.
- Giglio, S., Xiu, D. (2021). Asset pricing with omitted factors. *Journal of Political Economy*, 129(7), 1947–1990.
- Hansen, L.P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica*, 50(4), 1029–1054.
- Hansen, L.P., Jagannathan, R. (1997). Assessing specification errors in stochastic discount factor models. *Journal of Finance*, 52(2), 557–590.
- Härdle, W.K., Simar, L. (2019). Applied Multivariate Statistical Analysis, 5th ed. Springer.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.

References (4/6)

- Hou, K., Xue, C., Zhang, L. (2015). Digesting anomalies: an investment approach. *Review of Financial Studies*, 28(3), 650–705.
- Hou, K., Xue, C., Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5), 2019–2133.
- Jagannathan, R., Wang, Z. (1998). An asymptotic theory for estimating beta-pricing models using cross-sectional regression. *Journal of Finance*, 53(4), 1285–1309.
- Jegadeesh, N., Titman, S. (1993). Returns to buying winners and selling losers: implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- Jensen, M.C. (1968). The performance of mutual funds in the period 1945–1964. *Journal of Finance*, 23(2), 389–416.
- Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). Is there a replication crisis in finance?. *Journal of Finance*, 78(5), 2465–2518.
- Jheng, S.-L., Găman, Ș., Conda, A., Pele, D.T., Härdle, W.K., Teng, H.-W., Osterrieder, J. (2026). Crypto currency returns. SSRN Working Paper.
- Kaiser, H.F. (1960). The application of electronic computers to factor analysis. *Educational and Psychological Measurement*, 20(1), 141–151.
- Kan, R., Robotti, C., Shanken, J. (2013). Pricing model performance and the two-pass cross-sectional regression methodology. *Journal of Finance*, 68(6), 2617–2649.
- Kan, R., Zhang, C. (1999). Two-pass tests of asset pricing models with useless factors. *Journal of Finance*, 54(1), 203–235.
- Kelly, B.T., Pruitt, S., Su, Y. (2019). Characteristics are covariances: a unified model of risk and return. *Journal of Financial Economics*, 134(3), 501–524.

References (5/6)

- Kleibergen, F. (2009). Tests of risk premia in linear factor models. *Journal of Econometrics*, 149(2), 149–173.
- Kozak, S., Nagel, S., Santosh, S. (2020). Shrinking the cross-section. *Journal of Financial Economics*, 135(2), 271–292.
- Lettau, M., Pelger, M. (2020). Factors that fit the time series and cross-section of stock returns. *Review of Financial Studies*, 33(5), 2274–2325.
- Lewellen, J., Nagel, S., Shanken, J. (2010). A skeptical appraisal of asset pricing tests. *Journal of Financial Economics*, 96(2), 175–194.
- Lintner, J. (1965). The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics*, 47(1), 13–37.
- Liu, Y., Tsyvinski, A., Wu, X. (2022). Common risk factors in cryptocurrency. *Journal of Finance*, 77(2), 1133–1177.
- Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- McLean, R.D., Pontiff, J. (2016). Does academic research destroy stock return predictability?. *Journal of Finance*, 71(1), 5–32.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Newey, W.K., West, K.D. (1994). Automatic lag selection in covariance matrix estimation. *Review of Economic Studies*, 61(4), 631–653.
- Onatski, A. (2010). Determining the number of factors from empirical distribution of eigenvalues. *Review of Economics and Statistics*, 92(4), 1004–1016.

References (6/6)

- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *The European Journal of Finance*, 29(10), 1064–1105.
- Pesaran, M.H., Yamagata, T. (2024). Testing for alpha in linear factor pricing models with a large number of securities. *Journal of Financial Econometrics*, 22(2), 407–460.
- Politis, D.N., Romano, J.P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- Roll, R. (1977). A critique of the asset pricing theory's tests. Part I. *Journal of Financial Economics*, 4(2), 129–176.
- Ross, S.A. (1976). The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13(3), 341–360.
- Shanken, J. (1992). On the estimation of beta-pricing models. *Review of Financial Studies*, 5(1), 1–33.
- Sharpe, W.F. (1964). Capital asset prices: a theory of market equilibrium under conditions of risk. *Journal of Finance*, 19(3), 425–442.
- Vasicek, O.A. (1973). A note on using cross-sectional information in Bayesian estimation of security betas. *Journal of Finance*, 28(5), 1233–1239.
- White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4), 817–838.
- Data: Kenneth R. French Data Library, mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.